MCAMamba: Multilevel Cross-Modal Attention-Guided State-Space Model for Multisource Remote Sensing Image Classification

Mingyu Dou, Shi Qiu, Ming Hu, Xiaozhen Qiao, Huping Ye, Xiaohan Liao, *Member, IEEE*, and Zhe Sun

***Abstract*—Effective fusion of multisource remote sensing data remains a fundamental challenge for Earth observation, as convolutional neural network (CNN) and transformer models suffer from limited receptive fields and high computational complexity. While state-space models (SSMs) like Mamba show promise in sequence modeling, they face three critical challenges in multisource remote sensing: insufficient spatial–spectral coordination, cross-modal heterogeneity, and inadequate multiscale feature integration. To address these limitations, this article proposes a multilevel cross-modal attention-guided Mamba (MCAMamba) framework for joint classification of hyperspectral images (HSI) and light detection and ranging (LiDAR)/synthetic aperture radar (SAR) data. MCAMamba introduces a novel three-stage feature fusion pipeline: 1) the feature extraction attention (FExt-Attention) module enhances spatial structure and spectral information through parallel spatial-channel attention (Cha-Attn) mechanisms; 2) the SSM-Attention module achieves deep cross-modal fusion by combining attention mechanisms with SSM for parametric interaction; and 3) the feature fusion attention (FFus-Attention) module performs adaptive multiscale feature integration through global context modeling and cascaded attention. This hierarchical design enables superior feature representation with enhanced computational efficiency. Experiments on four public benchmark datasets (Houston2013, Houston2018, Augsburg, and Berlin) show that MCAMamba achieves overall accuracy (OA) of 94.75%, 93.35%, 92.46%, and 79.18%. The code will be available at https://github.com/Dmygithub/MCAMamba**

***Index Terms*—Attention mechanism, hyperspectral image (HSI), light detection and ranging (LiDAR), multisource remote sensing classification, state-space model (SSM), synthetic aperture radar (SAR).**

Received 30 May 2025; revised 29 August 2025; accepted 1 October 2025. Date of publication 6 October 2025; date of current version 29 October 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2023YFB3905700, in part by Shaanxi Key Research and Development Plan under Grant 2024SF-YBXM-678, and in part by the Western Light of China under Grant XAB2022YN10. *(Mingyu Dou, Shi Qiu, and Zhe Sun contributed equally to this work.) (Corresponding authors: Shi Qiu; Zhe Sun.)*

Mingyu Dou is with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China, and also with the University of Chinese Academy of Sciences, Beijing 100408, China (e-mail: doumingyu24@mails.ucas.ac.cn).

Shi Qiu and Ming Hu are with the Key Laboratory of Spectral Imaging Technology CAS, Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an 710119, China (e-mail: qiushi@opt.ac.cn; huming708@gmail.com).

Xiaozhen Qiao is with the School of Information Science and Technology, University of Science and Technology of China, Hefei 230026, China (e-mail: xiaozhennnqiao@mail.ustc.edu.cn).

Huping Ye is with the State Key Laboratory of Resources and Environment Information System, Institute of Geographic Sciences and Natural Resources Research, and the Key Laboratory of Low Altitude Geographic Information and Air Route, Civil Aviation Administration of China, Chinese Academy of Sciences, Beijing 100101, China (e-mail: yehp@igsnrr.ac.cn).

Xiaohan Liao is with the State Key Laboratory of Resources and Environment Information System, Institute of Geographic Sciences and Natural Resources Research, the Key Laboratory of Low Altitude Geographic Information and Air Route, Civil Aviation Administration of China, and the Research Center for UAV Applications and Regulation, Chinese Academy of Sciences, Beijing 100101, China (e-mail: liaoxh@igsnrr.ac.cn).

Zhe Sun is with the School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: sunzhe@nwpu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3618301

I. Introduction

With the rapid development of remote sensing technology, multisource remote sensing data have become a crucial foundation for Earth observation. Heterogeneous data collected through platforms such as satellites, aircraft, and uncrewed aerial vehicles provide abundant information sources for complex analytical tasks [1], [2]. These data have been successfully applied in critical fields including land cover classification [3], mineral resource detection [4], environmental monitoring [5], urban planning [6], and agricultural management [7]. As a core task in remote sensing applications, land use and land cover classification directly impacts crucial decision-making processes in environmental monitoring, resource management, and urban development [8], [9]. Therefore, efficient classification of multisource remote sensing data has become a pressing research focus.

Among the various remote sensing data sources, hyperspectral imagery, light detection and ranging (LiDAR), and synthetic aperture radar (SAR) are three key data sources. As illustrated in Fig. 1(a) and (d), hyperspectral images (HSI) offers unique advantages in material identification owing to its fine spectral features [10], [11]. LiDAR data [as shown in Fig. 1(b)] can accurately reflect the 3-D structure of the Earth's surface, providing crucial elevation information [12], [13]. Furthermore, SAR data [as shown in Fig. 1(e)], with its all-weather, all-day imaging capability, and penetration properties, can reliably provide surface texture and backscattering information [14], [15]. However, single data sources struggle to address complex scenes independently: HSI is susceptible to weather conditions, LiDAR has limitations in spectral

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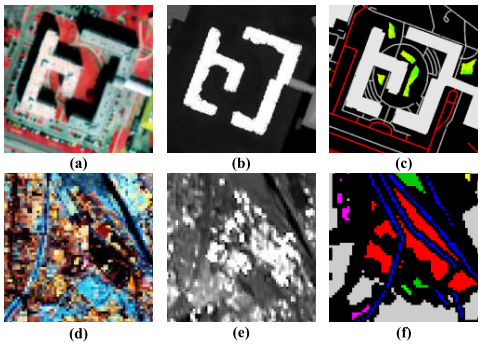


Fig. 1. (a)–(c) Houston2018 dataset: HSI, LiDAR elevation data, and ground truth land cover map. (d)–(f) Augsburg dataset: HSI, SAR, and ground truth land cover map.

perception, and SAR faces challenges with complex image textures and difficult interpretation. Consequently, integrating the complementary characteristics of multisource data to achieve superior classification results, exemplified by the accurate ground truth maps in Fig. 1(c) and (f), thereby enhancing classification accuracy and robustness, has become one of the core research directions in remote sensing applications [16], [17].

The rapid progress of deep learning has significantly advanced multisource remote sensing data classification. Convolutional neural networks (CNNs) [18], [19] offer strong feature extraction capabilities. However, their limited ability to capture long-range dependencies restricts the effective utilization of the rich spectral information present in hyperspectral data [20], [21]. To address this, transformers [22], [23] have been introduced. By leveraging self-attention mechanisms, they establish global dependencies and further enhance classification accuracy. Nevertheless, when applied to high-dimensional remote sensing data, transformers still face significant challenges. High computational complexity and low efficiency remain major bottlenecks, motivating the exploration of alternative model architectures.

Effective feature enhancement and fusion are also crucial for high-performance deep learning-based remote sensing classification. To this end, researchers have developed various techniques to achieve this goal. For instance, spatial attention (Spa-Attn) mechanisms adaptively focus on key spatial regions of ground objects, thereby strengthening spatial information representation [24]. Channel attention (Cha-Attn) dynamically adjusts the weights of different bands, improving the discriminative ability for core features [25], [26]. For multimodal data interaction, cross-modal attention (Cross-Attn) establishes information bridges between different data modalities [27]. Additionally, global context attention (GCon-Attn) enhances overall feature representation by capturing long-range dependencies between pixels [28]. Despite these advances, current methods still face challenges in computational efficiency and comprehensive sequential context modeling, especially when processing high-dimensional and multimodal remote sensing data. This highlights the need for more efficient sequence modeling approaches.

Recently, state-space models (SSMs) [29] and their variant Mamba [30], with their linear computational complexity and long-range modeling capabilities, have provided a new technological paradigm for multisource remote sensing classification. However, applying Mamba to joint classification of HSI and LiDAR/SAR still faces three key challenges.

1) *Insufficient coordination of spatial-channel information:* Although Mamba excels at modeling long-range dependencies in sequences, its native mechanism struggles to effectively coordinate spatial structural features (such as object boundaries and textures) with channel-wise information (such as spectral features and band correlations) in remote sensing applications. For example, road recognition requires simultaneous consideration of spatial morphology and material reflectance, but existing Mamba models lack a synergistic design for this [31], [32].
2) *Heterogeneity of modal features:* Mamba is primarily designed for single-modal information processing and struggles to effectively fuse the inherent feature heterogeneity of multisource remote sensing data. The feature representation of the same ground object varies significantly across different sensors (e.g., spectral information from HSI, elevation from LiDAR, and scattering mechanisms from SAR). This expressive gap between modalities poses a severe challenge for Mamba to achieve efficient cross-modal feature integration [33], [34].
3) *Limited integration of multiscale features:* Existing Mamba models lack specialized multiscale processing mechanisms, making it difficult to effectively integrate features of objects at different scales (such as buildings and vehicles). Furthermore, they have limitations in fine-grained local feature extraction and adaptive integration of multiscale information, hindering the comprehensive characterization and utilization of complex multilevel features in remote sensing data [35], [36].

To address these challenges, this article proposes the multilevel Cross-Attn-guided Mamba (MCAMamba) model, innovatively designing a systematic framework of attention-guided SSM to provide an efficient solution for HSI and LiDAR/SAR multisource remote sensing classification. Unlike existing methods that simply combine attention and SSM, MCAMamba designs a three-level feature fusion architecture: 1) feature extraction attention (FExt-Attention) module: collaboratively extracts and enhances spectral information of HSI and structural features of LiDAR/SAR through parallel spatial-Cha-Attn integrated in convolutional layers; 2) SSM-Attention module: integrates Cross-Attn with a SSM to promote deep interaction between heterogeneous data; and 3) feature fusion attention (FFus-Attention) module: achieves efficient adaptive fusion of multimodal features through uniquely integrating GCon-Attn, cascaded spatial-Cha-Attn, and feature enhancement mechanisms. This article makes the following main contributions.

1) Propose a novel multilevel attention-guided SSM framework, constructing an end-to-end pipeline from single-modal enhancement, cross-modal interaction, to multiscale fusion.

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2) Address insufficient spatial-channel coordination by designing differentiated attention strategies: the FExt-Attention module uses a parallel structure for comprehensive initial feature extraction, while the FFus-Attention module adopts a cascaded structure for precise final feature fusion
3) Develop the SSM-Attention module to tackle modal feature heterogeneity, combining Cross-Attn with Mamba and introducing a parametric interaction mechanism for efficient bidirectional fusion of HSI and LiDAR/SAR features.
4) Construct the FFus-Attention module to overcome multiscale integration limitations, adaptively perceiving and fusing features at different scales through global context and multilevel attention.

The organization of this article is as follows. Section II provides a comprehensive review of related work on multisource remote sensing classification, Mamba models, and attention mechanisms. Section III details the architectural design and key modules of the proposed MCAMamba framework. Section IV presents experimental results and offers an in-depth analysis based on four benchmark datasets. Finally, Section V concludes this article and discusses potential directions for future research.

II. Related Works

A. Multisource Remote Sensing Data Classification Methods

Early multisource remote sensing data classification primarily relied on traditional machine-learning methods such as support vector machines (SVMs) [37] and random forests (RFs) [38], to achieve multisource data fusion. These methods have relatively high computational efficiency, but they have inherent limitations in the depth and breadth of feature expression, making it difficult to fully exploit the complex complementarity between multisource information, which constrains further improvements in classification performance.

Traditional machine-learning methods have limitations in the depth and breadth of feature representation, making it difficult to fully exploit complex spatial and spectral information in multisource remote sensing data. To address these deficiencies, CNNs, with their powerful local feature extraction capabilities and hierarchical representation advantages, have been widely applied in multisource remote sensing. Xu et al. [39] proposed a two-branch CNN, using a dual-channel CNN to extract spatial and spectral features from HSI and LiDAR separately. Mohla et al. [40] introduced feature fusion and extraction network (FusAtNet), utilizing self-attention to generate HSI attention maps and cross-attention to process spatial feature fusion. Fang et al. [41] presented spatial-spectral cross-modal enhancement network (S^2ENet), designing spatial–spectral enhancement modules for multimodal data information interaction. Gao et al. [42] proposed the depthwise feature interaction network, which extracts complementary information through depthwise separable cross-attention modules. Li et al. [43] developed the asymmetric feature fusion network, employing weight-shared residual blocks and feature calibration modules to achieve feature modeling in the spatial dimension. Gao et al. [44] proposed prototype-based information compensation network, utilizing prototype learning to enhance feature representation in multisource data fusion.

Despite CNN's excellent performance in local feature perception, its inherent convolutional operations limit its ability to capture long-range dependencies and global contextual information, which is crucial for a comprehensive understanding of large-scale object relationships in remote sensing images. To break through this bottleneck, transformer models [45] based on self-attention mechanisms, with their outstanding global modeling capabilities, began to be introduced by researchers into multisource remote sensing classification. Yao et al. [46] proposed extended vision transformer (ExViT), utilizing position-shared transformer and separable convolution in parallel branches to process multimodal images, achieving multiscale feature extraction and fusion. Zhao et al. [47] introduced hierarchical CNN and transformer (HCT), realizing cross-modal feature fusion through cross-token attention, combining CNN's local feature extraction with transformer's global modeling advantages. Li et al. [28] proposed mixing self-attention and convolution network (MACN), integrating self-attention with convolution operations to achieve local and global multiscale feature perception. Lin et al. [48] introduced spatial–spectral masked autoencoder, leveraging self-supervised learning for robust feature extraction from multisource data. However, although transformer-based explorations significantly enhanced global feature representation capabilities, their core self-attention mechanism still faces the problem of quadratic computational complexity, necessitating more optimized sequence modeling approaches.

B. SSMs and Mamba

Addressing the computational efficiency bottleneck of transformers, SSMs [49], [50] with their efficient sequence modeling capabilities have received widespread attention as an emerging paradigm. Researchers like Smith et al. [29] enhanced SSM computational efficiency through techniques such as parallel scanning. The Mamba model [30], by introducing selective state-space mechanisms and input-dependent parameterization methods, achieved dynamic adjustment of the SSM state evolution process. This core innovation enables Mamba to effectively capture long-range dependencies while maintaining linear computational complexity, demonstrating enormous potential for remote sensing applications. In light of this, researchers have begun introducing it into remote sensing image processing: Ma et al. [51] applied Mamba to semantic segmentation, constructing a dual-branch architecture that fuses CNN local perception with Mamba global modeling; Gao et al. [52] proposed multiscale feature fusion Mamba (MSFMamba), focusing on enhancing multiscale feature integration and efficient sequence processing capabilities; while Cao et al. [53] explored integrating Cross-Attn into Mamba to promote deep fusion of multisource data. These cutting-edge explorations collectively confirm that Mamba and its variants, with their computational efficiency and sequence modeling advantages, are becoming a key technological pathway for multisource remote sensing data processing.

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Despite Mamba's advantages in sequence modeling, when directly applied to multisource remote sensing image classification tasks, it still faces three major challenges: insufficient spatial-channel information coordination, modal feature heterogeneity, and limited multiscale feature integration.

C. Multisource Remote Sensing Feature Enhancement and Fusion Techniques

To fully leverage the potential of Mamba in multisource remote sensing data processing, it is crucial to draw upon and develop mature feature enhancement and fusion techniques, particularly attention mechanisms. At the unimodal feature optimization level, Spa-Attn mechanisms [54] strengthen structural information modeling by focusing on key spatial regions, while Cha-Attn mechanisms [55] dynamically adjust channel weights to enhance the discriminative power of core information. Researchers have also explored cascaded attention architectures like CBAM [56] or parallel attention architectures as proposed by Dou et al. [57] to synergistically enhance spatial and channel features.

At the multimodal data interaction level, Cross-Attn mechanisms [58], [59] build information bridges between different data sources, effectively promoting the deep fusion of heterogeneous features. Wang et al. [60] developed a nearest neighbor-based contrastive learning framework, employing contrastive learning strategies to enhance feature discrimination in HSI and LiDAR classification. To improve the model's understanding of the global scene, GCon-Attn mechanisms [61] enhance global feature expression by capturing long-range dependencies and overall semantic information. Gao et al. [62] proposed an adaptive frequency enhancement network that integrates spatial-frequency features for enhanced global context modeling. Li et al. [28] balanced global and local information modeling by integrating self-attention with convolution. Furthermore, broader fusion strategies, such as multiscale feature fusion networks [63], have significantly improved classification performance in complex scenes by integrating multilevel semantic information.

Although existing methods have made significant progress, Mamba still faces challenges such as insufficient spatial-channel information coordination, modal feature heterogeneity, and limited multiscale feature integration when applied to multisource remote sensing classification. The MCAMamba architecture proposed in this article, by constructing an attention-guided state-space model framework, organically combines spatial, channel, cross-modal, and GCon-Attn with the Mamba model. This aims to achieve a complete processing pipeline from unimodal feature enhancement and heterogeneous data interaction to multiscale information fusion, providing a new approach for multisource remote sensing data classification.

III. Methodology

This section first introduces the basic concepts and mathematical representations of SSMs and feature enhancement techniques, followed by an exposition of the MCAMamba architecture design, sequentially describing each functional module in the model's three-level fusion strategy and their implementation details.

A. Preliminaries

1) State-Space Models: SSMs originate from control theory, mapping an input sequence $x = (x_1, x_2, \ldots, x_L)$ to an output sequence $y = (y_1, y_2, \ldots, y_L)$. Its basic form can be represented as a linear ordinary differential equation

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \quad y(t) = \mathbf{C}h(t) + \mathbf{D}x(t) \tag{1}$$

where $h(t) \in \mathbb{R}^N$ is the system's hidden state, $\mathbf{A} \in \mathbb{R}^{N\times N}$ controls state evolution, $\mathbf{B} \in \mathbb{R}^{N\times D_{\text{in}}}$ and $\mathbf{C} \in \mathbb{R}^{D_{\text{out}}\times N}$ are projection matrices, and $\mathbf{D} \in \mathbb{R}^{D_{\text{out}}\times D_{\text{in}}}$ provides a direct pathway.

In deep learning applications, continuous SSMs need to be discretized. Using the zero-order hold (ZOH) method and introducing a time step Δ, the discretized state equation becomes

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t, \quad y_t = \mathbf{C}h_t + \mathbf{D}x_t \tag{2}$$

where $\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\overline{\mathbf{B}} \approx \Delta\mathbf{B}$.

Traditional SSMs cannot dynamically adjust parameters based on input, whereas Mamba introduces an input-dependent parameter generation mechanism

$$\mathbf{B}_t = f_B(x_t), \quad \mathbf{C}_t = f_C(x_t), \quad \Delta_t = f_\Delta(x_t). \tag{3}$$

2) Feature Enhancement Techniques: Feature enhancement techniques are crucial for improving model representation capabilities in deep learning, particularly effective for multisource remote sensing data processing. This article employs four attention mechanisms to achieve feature enhancement:

Cha-Attn adaptively adjusts feature channel weights to highlight the most discriminative channel features in hyperspectral or multimodal data

$$A_c = \sigma\left(\text{MLP}\left(\text{AvgPool}(F) + \text{MaxPool}(F)\right)\right) \tag{4}$$

where σ is the sigmoid activation function, AvgPool and MaxPool represent average pooling and maximum pooling operations on the input feature F, respectively, and MLP denotes a multilayer perceptron.

Spa-Attn adaptively focuses on key spatial regions in feature maps to capture land cover boundaries and structural features in remote sensing images

$$A_s = \sigma\left(\text{Conv}\left(\left[\text{AvgPool}(F); \text{MaxPool}(F)\right], W_s\right)\right) \tag{5}$$

where $[\cdot;\cdot]$ represents channel-wise concatenation, AvgPool and MaxPool indicate pooling operations along the channel dimension, and W_s represents convolution weights.

GCon-Attn acquires global semantic information from feature maps, enhancing the ability to recognize large-scale land objects

$$A_g = \sigma\left(\text{MLP}\left(\text{GlobalAvgPool}(F)\right)\right). \tag{6}$$

Cross-Attn enables information interaction between different modality data, used in this work to promote deep fusion of HSI and LiDAR/SAR data

$$\text{CrossAttn}(\mathbf{X}_h, \mathbf{X}_r) = \mathbf{X}_r \odot \sigma\left(\text{Conv}\left(\text{Cat}\left[\mathbf{X}_h, \mathbf{X}_r\right]\right)\right) \tag{7}$$

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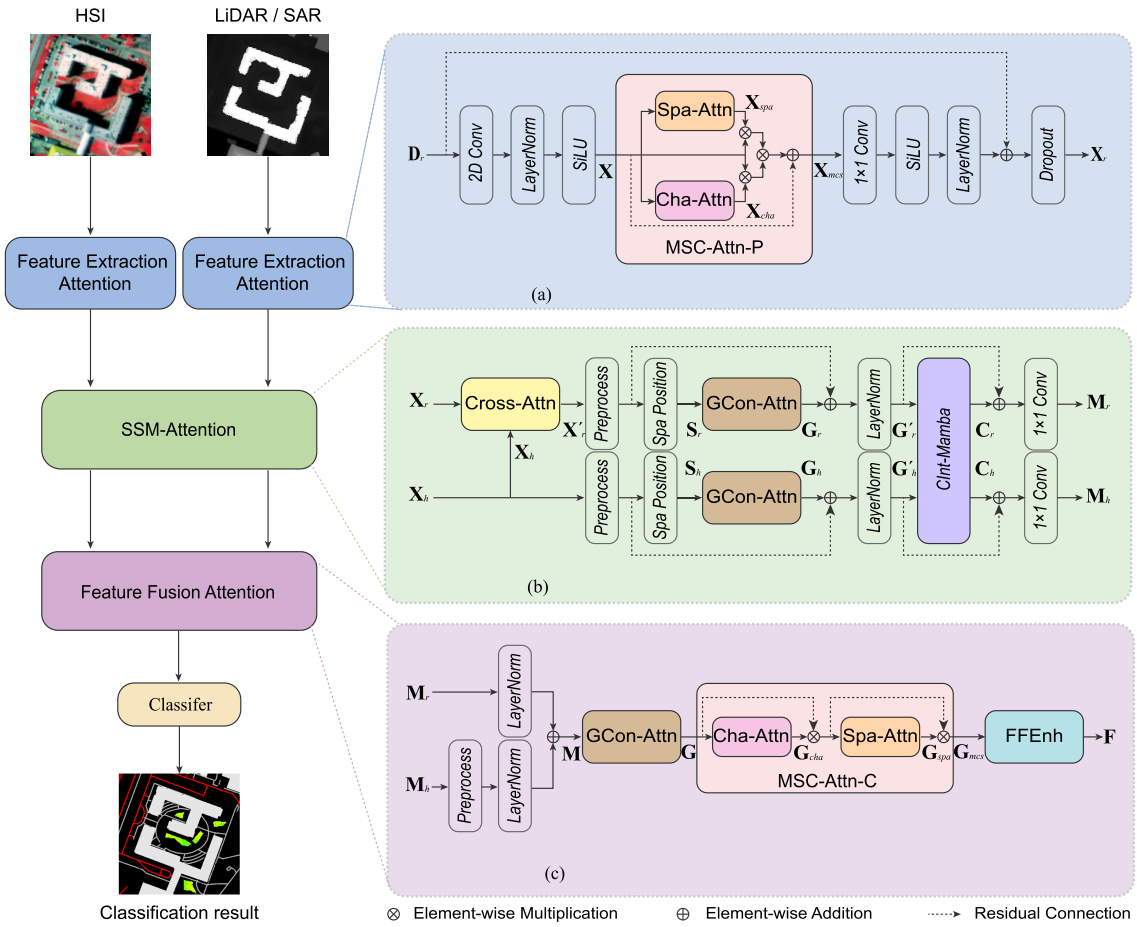


Fig. 2. Proposed MCAMamba architecture for HSI and LiDAR/SAR data classification. The model establishes an end-to-end three-level feature fusion framework from single-modal feature extraction, heterogeneous data interaction, to multiscale feature fusion. (a) FExt-Attention module employs parallel spatial-Cha-Attn blocks for single-modal feature extraction and enhancement. (b) SSM-attention module integrates a Cross-Attn block with a cross-modal Mamba model to achieve deep interaction between heterogeneous data. (c) FFus-Attention module combines a GCon-Attn block with cascaded spatial-Cha-Attn blocks to accomplish adaptive fusion of multiscale features.

where Cat$[\cdot,\cdot]$ represents feature concatenation, Conv denotes a mapping network containing convolution layers, activation functions, and normalization, and $\odot$ represents element-wise multiplication. This design establishes information pathways between different modalities.

B. Overall Framework of the Proposed MCAMamba

As shown in Fig. 2, MCAMamba constructs a systematic framework of attention-guided SSM, realizing end-to-end processing from feature extraction, cross-modal interaction to multiscale fusion. The architecture is designed based on a three-level fusion strategy, including the FExt-Attention module as shown in Fig. 2(a), the SSM-Attention module as shown in Fig. 2(b), and the FFus-Attention module as shown in Fig. 2(c). The preprocessed HSI and LiDAR/SAR data, serving as parallel input branches, first undergo single-modal feature extraction and enhancement through the FExt-Attention module, then achieve deep interaction of heterogeneous data and sequence modeling via the SSM-Attention module, and finally complete adaptive fusion of multiscale features in the FFus-Attention module, with pixel-level classification results output by a fully connected classifier. The detailed design of each functional module will be elaborated sequentially in Sections III-C and III-E.

C. FExt-Attention Module

As shown in Fig. 2(a), the FExt-Attention module (using the LiDAR/SAR branch as an example) first processes the feature $\mathbf{D}_r$ through convolution, followed by LayerNorm and SiLU activation to obtain $\mathbf{X}_{\text{in}}$. It is important to note that while both HSI and LiDAR/SAR data enter the FExt-Attention module with $[B, C, H, W]$ dimensions, they exhibit distinct data characteristics: HSI's C dimension represents PCA-reduced spectral bands, whereas LiDAR/SAR's C dimension contains elevation or polarimetric features. Consequently, different convolution strategies are employed for initial feature extraction. The diagram illustrates the LiDAR/SAR branch utilizing 2-D convolution, while the HSI branch adopts 3-D convolution to effectively capture spectral-spatial relationships. The strategic placement of convolution before LayerNorm ensures spatial feature extraction precedes normalization for optimal atten tion mechanism performance. It then inputs this into the multiscale spatial-Cha-Attn module MSC-Attn-P (Fig. 3) for feature enhancement, outputting $\mathbf{X}_{\text{msc}}$. $\mathbf{X}_{\text{msc}}$ and $\mathbf{X}_{\text{in}}$ are fused

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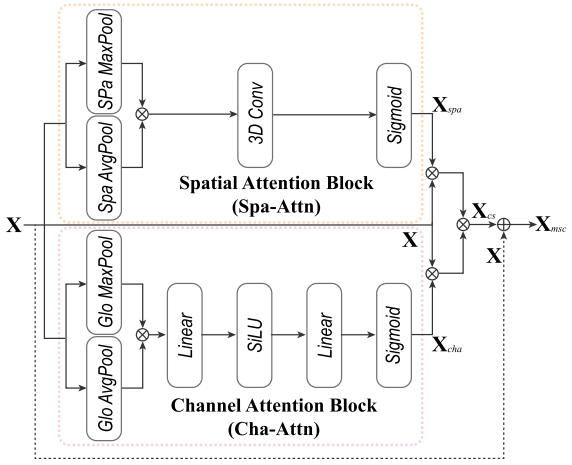


Fig. 3. Multiscale spatial-Cha-Attn block with parallel design (MSC-Attn-P). The orange area represents the Spa-Attn block. The pink area shows the Cha-Attn block.

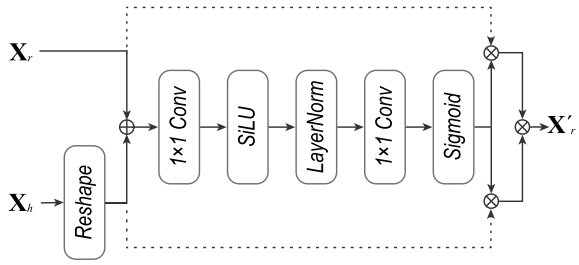


Fig. 4. Structure of the Cross-Attn block. This module enables the modulation of LiDAR/SAR features $\mathbf{X}_r$ by HSI features $\mathbf{X}_h$.

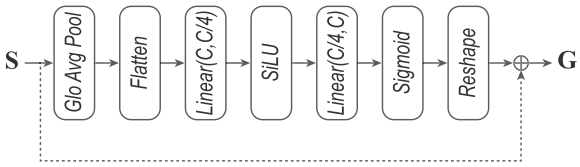


Fig. 5. Structure of the GCon-Attn block. This module captures global contextual information from input features to enhance the model's ability to perceive large-scale spatial structures.

through a residual connection, followed by 1×1 convolution, SiLU activation, LayerNorm, and dropout processing to finally output $\mathbf{X}_{\text{msc}}$. MCAMamba designs two parallel branches for HSI and LiDAR/SAR, with the only difference being in convolution dimensions: the HSI branch employs 3-D convolution, while the LiDAR/SAR branch uses 2-D convolution.

Unlike the cascaded approach in CBAM [56], MSC-Attn-P adopts a parallel design for spatial and Cha-Attn, simultaneously enhancing spatial structure and channel information. In MSC-Attn-P, the Spa-Attn module performs average and maximum pooling on the input $\mathbf{X}$, respectively, concatenates them, and obtains spatial weights through a convolution layer and Sigmoid function

$$\mathbf{A}_s = \sigma\left(\text{Conv}\left(\left[\text{AvgPool}\left(\mathbf{X}\right); \text{MaxPool}\left(\mathbf{X}\right)\right], W_s\right)\right). \tag{8}$$

The enhanced feature is $\mathbf{X}_{\text{spa}} = \mathbf{X} \odot \mathbf{A}_s$. Simultaneously, the Cha-Attn module performs global average and maximum pooling, adds the results, and feeds them into an MLP with a bottleneck structure and a Sigmoid function to generate channel weights

$$\mathbf{A}_c = \sigma\left(\text{MLP}\left(\text{GlobalAvgPool}\left(\mathbf{X}\right) + \text{GlobalMaxPool}\left(\mathbf{X}\right)\right)\right) \tag{9}$$

where the MLP structure is $W_2\delta(W_1 x)$. The enhanced feature is $\mathbf{X}_{\text{cha}} = \mathbf{X} \odot \mathbf{A}_c$. Finally, the spatial and channel enhanced features are multiplied and added to the original input through a residual connection to obtain the enhanced feature

$$\mathbf{X}_{\text{msc}} = \mathbf{X}_{\text{spa}} \odot \mathbf{X}_{\text{cha}} + \mathbf{X}. \tag{10}$$

Subsequent experiments will use t-SNE visualization to verify the effectiveness of this parallel attention for single-modal feature extraction.

Following this processing, the enhanced HSI and LiDAR/SAR features are denoted as $\mathbf{X}_h$ and $\mathbf{X}_r$, respectively, which serve as inputs to the SSM-Attention module for cross-modal interaction.

D. SSM-Attention Module

To address the challenges of modal feature heterogeneity and multiscale integration in multisource remote sensing data processing, the proposed SSM-Attention module [Fig. 2(b)] integrates Cross-Attn, GCon-Attn, and cross-modal interaction Mamba (CInt-Mamba). This module achieves heterogeneous data sequence modeling and feature fusion through the following core components:

1) Cross-Modal Attention: Inspired by [64], we designed the Cross-Attn module (Fig. 4) for dynamic interaction between HSI and LiDAR/SAR features. This module receives FExt-Attention-enhanced HSI features $\mathbf{X}_h$ and LIDAR/SAR features $\mathbf{X}_r$, first reshaping the HSI features to align with the radar feature spatial dimensions. Its core computation is

$$\mathbf{X}_r' = \mathbf{X}_r \odot \sigma\left(\text{Conv}\left(\text{Cat}\left[\text{Reshape}\left(\mathbf{X}_h\right), \mathbf{X}_r\right]\right)\right). \tag{11}$$

Through this mechanism, HSI information guides radar feature enhancement, outputting the modulated $\mathbf{X}_r'$, while $\mathbf{X}_h$ is directly passed to the next stage.

2) Global Context Attention: As shown in Fig. 5, this module is designed based on GCNet [65], extracting global descriptors of feature $\mathbf{S}$ through global average pooling

$$\mathbf{f} = \text{Flatten}\left(\text{GlobalAveragePool}\left(\mathbf{S}\right)\right). \tag{12}$$

Subsequently, global context weights $\mathbf{A}_g$ are generated through an MLP bottleneck structure and Sigmoid function

$$\mathbf{A}_g = \sigma\left(\text{Linear}_2\left(\text{SiLU}\left(\text{Linear}_1\left(\mathbf{f}\right)\right)\right)\right). \tag{13}$$

These weights act on the original input features to enhance global contextual information

$$\mathbf{G} = \mathbf{S} \odot \text{reshape}\left(\mathbf{A}_g, B, C, 1, 1\right). \tag{14}$$

This process is applied to both HSI and LiDAR/SAR feature branches, producing $\mathbf{G}_h$ and $\mathbf{G}_r$, enabling the model to adaptively focus on global information and enhance its ability to recognize complex land cover structures.

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Algorithm 1 CInt-SSM, as Shown in Fig. 6(b)

Input: $\mathbf{S}_r, \mathbf{S}_h : 0, 100, 0(\mathrm{P}, \mathrm{C}), (\mathrm{P}, \mathrm{C})$
Output: $\mathbf{Y}_r : 0, 100, 0(\mathrm{P}, \mathrm{C})$
/* Interaction Awareness Processing */
$\mathbf{I}_r, \mathbf{I}_h \leftarrow \mathbf{IA}(\mathbf{S}_r, \mathbf{S}_h)$
/* $\mathbf{I}_{h,r} = \sigma(\mathbf{W}[\mathbf{S}_h; \mathbf{S}_r])$ */
$\mathbf{x}_r : 0, 100, 0(\mathrm{P}, \mathrm{C}) \leftarrow \mathbf{S}_r \odot \mathbf{I}_h$
/* Fixed Parameter Acquisition */
$\mathbf{A}_r, \mathbf{D}_r : 0, 100, 0(\mathrm{C}, \mathrm{N}), (\mathrm{C}) \leftarrow \mathbf{ParamInit}()$
/* Cross-Dynamic Parameter Generation */
$[\mathbf{B}r, \mathbf{C}_h] : 0, 100, 0(\mathrm{P}, \mathrm{N}), (\mathrm{P}, \mathrm{N}) \leftarrow \mathbf{Linear}_1(\mathbf{S}_h)$
$\mathbf{\Delta}r : 0, 100, 0(\mathrm{P}, \mathrm{C}) \leftarrow \mathbf{Linear}_2(\mathbf{S}_h)$
/* State-Space Parameter Transformation */
$\overline{\mathbf{A}}_r \leftarrow \exp(\mathbf{\Delta}_r \mathbf{A}_r)$
$\overline{\mathbf{B}}_r \leftarrow \mathbf{\Delta}_r \mathbf{B}_r$
$\mathbf{h}_r^0 \leftarrow \mathbf{0}$
/* $\mathbf{h}_r^t = \overline{\mathbf{A}}_r \mathbf{h}_r^{t-1} + \overline{\mathbf{B}}_r \mathbf{x}_r^t$ */
/* $\mathbf{y}_r^t = \mathbf{C}_h \mathbf{h}_r^t + \mathbf{D}_r \mathbf{x}_r^t$ */
$\mathbf{Y}_r \leftarrow \mathrm{SSM}(\overline{\mathbf{A}}_r, \overline{\mathbf{B}}_r, \mathbf{C}_h, \mathbf{D}_r)(\mathbf{x}_r)$
return $\mathbf{Y}_r$

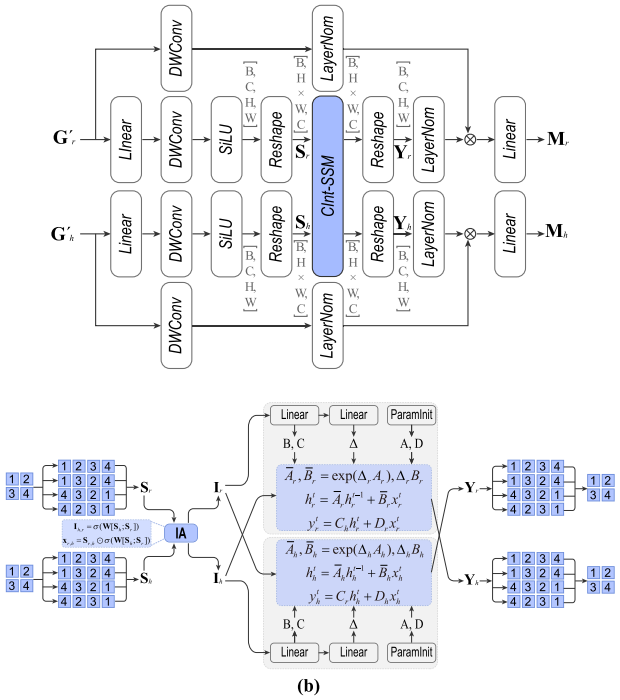


Fig. 6. (a) Structure of the CInt-Mamba module. (b) Structure of the cross-modal interaction SSM (CInt-SSM).

3) Cross-Modal Interaction Mamba: As shown in Fig. 6(a), our proposed CInt-Mamba module effectively addresses the issue of modal heterogeneity through bidirectional parameterized interaction and dynamic perception mechanisms. This module first processes the input globally context-enhanced features $\mathbf{G}_r'$ and $\mathbf{G}_h'$: they are, respectively, passed through linear layers, depthwise separable convolution (DWConv), and SiLU activation functions to generate intermediate representations with spatial dimensions $[B, C, H, W]$. Since Mamba requires 1-D sequence input while current features maintain 2-D spatial structure, spatial serialization is achieved through reshape operations. As illustrated in the diagram, 2-D spatial features with dimensions $[B, C, H, W]$ are reshaped into 1-D sequence format $[B, H \times W, C]$, where each spatial location (h, w) becomes a sequential token containing a C-dimensional feature vector. This transformation preserves complete spatial information while enabling sequential processing. The serialized features $\mathbf{S}_r$ and $\mathbf{S}_h$ are then fed as input to the core component, CInt-SSM, whose structure is depicted in Fig. 6(b). The computational process is detailed in Algorithm 1, exemplified by the LiDAR/SAR branch:

First, the interaction awareness (IA) mechanism is employed to compute interaction weights $\mathbf{I}_r$ and $\mathbf{I}_h$. Subsequently, these weights are applied to the original features $\mathbf{S}_r$ and $\mathbf{S}_h$ via element-wise multiplication, achieving adaptive modulation and semantic alignment of modal features. To achieve deeper cross-modal interaction, CInt-SSM adopts a dual-stage full-parameter cross-generation mechanism, distinguishing it from existing single-stage approaches: Sigma [66] employs single-parameter cross-modal generation (only C parameter), MSFMamba [52] primarily performs modal interaction at the feature fusion layer, while CInt-SSM establishes full-parameter cross-modal dependencies at the parameter generation stage and further strengthens interaction at the feature fusion stage. Specifically, HSI features ($\mathbf{S}_h$) generate the dynamic parameters ($\mathbf{B}_r$, $\mathbf{C}_h$, $\mathbf{\Delta}_r$) required for the LiDAR/SAR branch through linear transformation, while LiDAR/SAR features ($\mathbf{S}_r$) generate the corresponding dynamic parameters ($\mathbf{B}_h$, $\mathbf{C}_r$, $\mathbf{\Delta}_h$) for the HSI branch. These dynamically generated parameters, combined with preinitialized fixed parameters $\mathbf{A}_r$ and $\mathbf{D}_r$, undergo exponential and linear transformations to form the actual operational parameters $\overline{\mathbf{A}}_r$ and $\overline{\mathbf{B}}_r$ for the SSM.

In the SSM computation stage, this model employs a parameter cross-over mechanism: as shown in the blue modules in Fig. 6(b), the output $\mathbf{Y}_r$ of the LiDAR/SAR branch depends on the parameter $\mathbf{C}_h$ generated from HSI features, and vice versa. This dual-stage interaction design ensures full-process cross-modal information transfer from parameter generation to feature fusion, achieving more comprehensive modal collaborative modeling than single-stage methods. Following CInt-SSM processing, the output 1-D sequence features $\mathbf{Y}_r$ and $\mathbf{Y}_h$ are restored to spatial structure through inverse reshape operations, transforming from $[B, H \times W, C]$ back to $[B, C, H, W]$ format. Subsequent LayerNorm and linear transformations generate the final cross-modal features $\mathbf{M}_r$ and $\mathbf{M}_h$ enriched with intermodal interaction information.

SSM-Attention constructs a multilevel cross-modal interaction framework to address the challenges of modal heterogeneity and insufficient multiscale feature integration.

1) Cross-Attn achieves preliminary alignment and modulation at the feature level, mitigating representational differences between heterogeneous data.
2) GCon-Attn enhances the model's perception of large-scale spatial structures through global context modeling, providing global semantic guidance for subsequent fusion.
3) CInt-Mamba establishes a deeply interactive SSM, achieving comprehensive modal interaction modeling

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from feature modulation, parameter generation, to output mapping.

The output features from CInt-Mamba for HSI and LiDAR/SAR branches are denoted as $\mathbf{M}_h$ and $\mathbf{M}_r$, respectively, which contain deep cross-modal interactions and serve as inputs to the FFus-Attention module for final feature fusion.

E. FFus-Attention Module

In traditional multisource remote sensing classification, feature fusion often employs simple concatenation or weighted summation, making it difficult to achieve deep integration of information across modalities. The FFus-Attention module proposed in this article integrates GCon-Attn, cascaded spatial-Cha-Attn, and feature enhancement strategies to achieve efficient multimodal feature fusion.

As shown in Fig. 2(c), FFus-Attention receives HSI features $\mathbf{M}_h$ and radar features $\mathbf{M}_r$ from SSM-Attention, normalizes and concatenates them into feature $\mathbf{M}$, which then sequentially passes through three processing stages: GCon-Attn, cascaded multiscale spatial-Cha-Attn (MSC-Attn-C), and fusion feature enhancement (FFEnh) module.

The GCon-Attn module shares the same structure as previously described, capturing global semantic information of features through global average pooling and multilayer perceptrons to generate globally enhanced features $\mathbf{G}$. The MSC-Attn-C module performs refined processing on the features

$$\mathbf{A}_c = \sigma\left(\mathrm{MLP}\left(\mathrm{AvgPool}\left(\mathbf{G}\right) + \mathrm{MaxPool}\left(\mathbf{G}\right)\right)\right) \tag{15}$$

$$\mathbf{G}_{\mathrm{cha}} = \mathbf{G} \odot \mathbf{A}_c \tag{16}$$

$$\mathbf{F}_{\mathrm{cat}} = \left[\mathrm{AvgPool}\left(\mathbf{G}_{\mathrm{cha}}\right); \mathrm{MaxPool}\left(\mathbf{G}_{\mathrm{cha}}\right)\right] \tag{17}$$

$$\mathbf{A}_s = \sigma\left(\mathrm{Conv2D}\left(\mathbf{F}_{\mathrm{cat}}, W_s\right)\right) \tag{18}$$

$$\mathbf{G}_{\mathrm{mcs}} = \mathbf{G}_{\mathrm{cha}} \odot \mathbf{A}_s. \tag{19}$$

Unlike the parallel design shown in Fig. 3, MSC-Attn-C adopts a cascaded structure, first applying Cha-Attn followed by Spa-Attn, progressively focusing on key channels and spatial regions to provide more precise feature representations for complex land cover classification.

The feature $\mathbf{G}_{\mathrm{mcs}}$ output by the MSC-Attn-C module subsequently enters the FFEnh module for final feature optimization, as shown in the cyan module on the right side of Fig. 2(c). FFEnh, through cascaded operations of feature dimensionality reduction and nonlinear transformation, plays a crucial role in feature dimension coordination and nonlinear enhancement in the MCAMamba architecture, effectively improving the discriminative ability of the subsequent classifier. Its mathematical expression is

$$\mathbf{F} = \mathrm{Dropout}\left(\mathrm{SiLU}\left(\mathrm{LayerNorm}\left(\mathrm{Conv}_{1\times1}\left(\mathbf{G}_{\mathrm{mcs}}\right)\right)\right)\right). \tag{20}$$

Finally, the fused feature $\mathbf{F}$ is fed into a multilayer perceptron for classification: $\hat{y} = \mathrm{MLP}(\mathbf{F})$. Through the synergistic effect of global context awareness and local feature refinement, FFus-Attention effectively integrates complementary information from multimodal data, providing highly discriminative feature representations for classification.

IV. Experimental Results and Analysis

This section first introduces the experimental datasets and evaluation metrics, determines the optimal configuration through parameter sensitivity analysis, compares with advanced methods, verifies the contribution of each module and the rationality of the design through ablation experiments and architecture substitution, and finally conducts an in-depth analysis of key components.

A. Dataset Description

To validate the effectiveness and versatility of the MCA-Mamba model, we used four publicly available multimodal datasets: Houston2013, Houston2018, Augsburg, and Berlin. As shown in Fig. 7, the HSI, corresponding LiDAR/SAR data, and ground truth labels are displayed in pseudocolor. The first two datasets are used for joint classification of HSI and LiDAR data, while the latter two are used for joint classification of HSI and SAR data.

1) Houston2013 Dataset: This dataset originates from the 2013 IEEE Geoscience and Remote Sensing Society Data Fusion Contest, covering urban areas of Houston, Texas, and its surrounding regions [67]. The data were collected in June 2012, with a spatial resolution of 2.5 m, covering 755 × 349 pixels. The HSI data contain 144 spectral bands ranging from 380 to 1050 nm, while the LiDAR data provide elevation information in the form of a digital surface model (DSM). These two complementary data sources are jointly annotated with 15 different land cover types, providing a standard benchmark for multisource remote sensing analysis.

2) Houston2018 Dataset: This is part of the 2018 IEEE Geoscience and Remote Sensing Society Data Fusion Contest, mainly covering the University of Houston campus and adjacent urban areas [68]. The data was collected in February 2017, with HSI data at a spatial resolution of 1 m and LiDAR data at 0.5 m. The HSI data contain 48 spectral bands ranging from 380 to 1050 nm, while the LiDAR data is multispectral, operating at three wavelengths: 1550, 1064, and 532 nm, provided in DSM form. This dataset is annotated with 20 land cover types, offering richer object samples.

3) Augsburg Dataset: This dataset primarily covers the Augsburg region in Germany [69], consisting of hyperspectral data acquired by the HySpex sensor and SAR data collected by the Sentinel-1 sensor. Both data types have a spatial dimension of 332 × 485 pixels with a ground sampling distance of 30 m. The hyperspectral imagery contains 180 spectral bands ranging from 0.4 to 2.5 μm, while the SAR data provides four features obtained through polarimetric decomposition, including VV intensity, VH intensity, and the real and imaginary parts of the off-diagonal elements of the polarimetric SAR covariance matrix. The dataset includes seven land cover categories, presenting a typical mixed urban and suburban scene.

4) Berlin Dataset: This dataset covers urban and suburban areas of Berlin, Germany [70]. The hyperspectral data are based on the environmental mapping and analysis program (EnMAP) data simulated from the HyMap sensor, containing 244 spectral bands ranging from 400 to 2500 nm, with a spatial dimension of 797 × 220 pixels. The SAR data come from the

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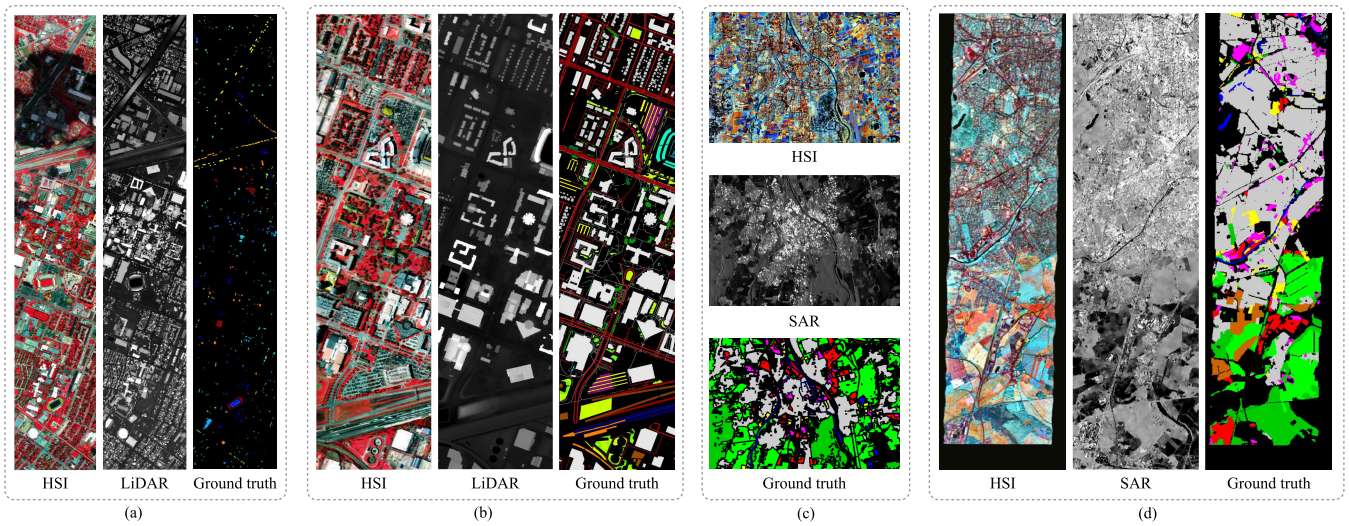


Fig. 7. Visualization of the four datasets used in our experiments. (a) Houston2013 dataset. (b) Houston2018 dataset. (c) Augsburg dataset. (d) Berlin dataset.

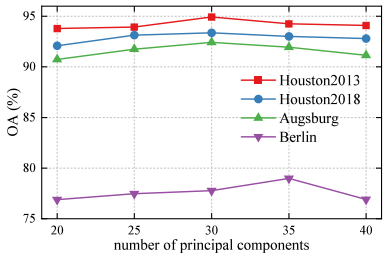


Fig. 8. Relationship between OA and the number of principal components.

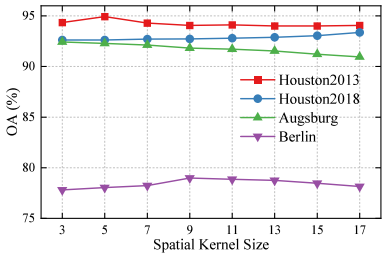


Fig. 9. Relationship between OA and the spatial kernel size.

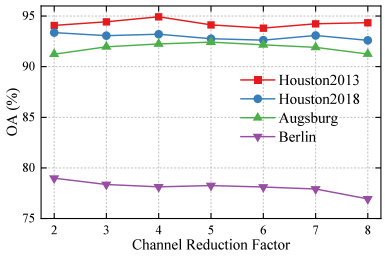


Fig. 10. Relationship between OA and the channel reduction factor.

European Space Agency's Sentinel-1 dual-polarization (VV-VH) single-look complex product, with dimensions of 1723 × 476 pixels, and spatial matching was achieved through nearest-neighbor interpolation. This dataset is annotated with eight urban land cover types, and its complex spatial structure provides an ideal testing scenario for algorithm evaluation.

This study employs three complementary evaluation metrics: overall accuracy (OA), average accuracy (AA), and kappa. OA represents the proportion of correctly classified samples to the total samples, intuitively reflecting the overall performance of the model. AA is the average of recall rates for each category, capable of evaluating the model's recognition capability for each class, especially when facing imbalanced datasets. Kappa reflects the consistency between predicted labels and true labels by comparing classification results with the possibility of random classification.

B. Hyperparameter Sensitivity Analysis

This section systematically analyzes five core parameters of the MCAMamba model, describing their impact on model performance. We follow a logical sequence from data preprocessing to feature extraction to model regularization, discussing how these parameters collectively influence model performance.

1) Number of Principal Components in HSI: When using PCA to reduce spectral redundancy in HSI, the number of principal components (N_p) is a key parameter determining the amount of information input to the model. We tested the effect of N_p ranging from 20 to 40. As shown in Fig. 8, when N_p increases from 20 to 30–35, the performance of all datasets significantly improves, with Houston2013, Houston2018, and Augsburg performing optimally at $N_p = 30$. The Berlin dataset reaches its peak at $N_p = 35$. When N_p exceeds 35, performance begins to decline slightly.

2) Spatial Kernel Size: When the model uses Spa-Attn, the size of the spatial convolution kernel affects its ability to capture spatial context information. We tested eight different spatial convolution kernel sizes, ranging from 3 to 17. As shown in Fig. 9, the OA reaches its highest when the Spatial Kernel Size is 5, 17, 3, and 9 for Houston2013, Houston2018, Augsburg, and Berlin, respectively.

3) Channel Reduction Factor: The channel reduction factor determines the compression degree of feature channels in the Cha-Attn module. As shown in Fig. 10, Houston2018 and Berlin perform best with a channel reduction factor of 2.

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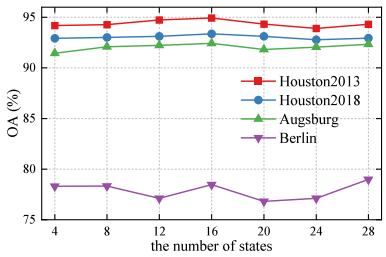


Fig. 11. Relationship between OA and the number of states in SSM.

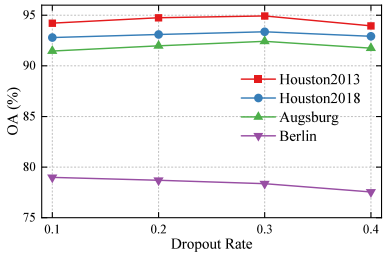


Fig. 12. Relationship between OA and the dropout rate.

Augsburg performs optimally with a reduction factor of 5. The Houston2013 dataset performs best with a reduction factor of 4.

4) Number of States: The number of states, as a core parameter of the Mamba SSM, determines the model's ability to capture sequence dependencies. We tested seven different numbers of states, ranging from 4 to 28. As shown in Fig. 11, Houston2013 and Houston2018 reach their performance peaks with 16 states. Augsburg and Berlin perform optimally with 28 states.

5) Dropout Rate: As an important regularization parameter, the dropout rate plays a crucial role in preventing model overfitting. We tested four dropout values ranging from 0.1 to 0.4. As shown in Fig. 12, the optimal dropout value for Houston2013, Houston2018, and Augsburg is 0.3. Berlin performs best with a dropout of 0.1.

The above experiments indicate significant differences in model parameters across different datasets, reflecting the influence of data characteristics on parameter sensitivity. Houston2013 and Houston2018 show significant differences in optimal spatial convolution kernel sizes, with Houston2013's optimal kernel being 5 while Houston2018's is 17, which is related to Houston2018's more complex spatial structure. For Augsburg and Berlin, the two HSI and SAR fusion datasets require a larger number of states (28) in the SSM parameters, possibly influenced by the complex coherent characteristics of SAR data. Additionally, the Berlin dataset shows unique performance in regularization intensity, with an optimal dropout value of only 0.1, lower than the 0.3 of other datasets, which is related to its category distribution characteristics. This indicates that model parameter configurations should be specifically optimized according to different application scenarios and data characteristics to fully leverage the advantages of multisource remote sensing data fusion.

Through the above parameter experiments, we determined the optimal configurations for MCAMamba on different datasets. The unified training configuration details are as follows: Adam optimizer with an initial learning rate of 1×10^{-4}, cosine annealing learning rate scheduler, 100 training epochs, and batch size of 128. During training, we employed random sampling and random flip data augmentation strategies. All experiments were implemented using the PyTorch framework in an environment equipped with an NVIDIA RTX 3090 GPU and 24 GB RAM.

C. Comparison With State-of-the-Art

We systematically compared the proposed MCAMamba with six state-of-the-art multisource remote sensing data classification methods, including FusAtNet [40], S^2ENet [41], ExViT [46], HCT [47], MACN [28], and multiscale feature Mamba (MSFMamba) [52].

To ensure fairness and reliability of the comparison, this study adopted the same data preprocessing pipeline, experimental configuration, and evaluation standards as the state-of-the-art model MSFMamba [52] from the same year, comparing evaluation metrics between MCAMamba and MSFMamba along with five other representative advanced methods to verify MCAMamba's performance advantages. The selected comparison models represent advanced models with different technical approaches in the field of multisource remote sensing classification: FusAtNet [40] employs a dual-track architecture to enhance spectral and spatial information separately with attention, achieving effective fusion of multisource data. S^2ENet [41] designed specialized modules to process information exchange between different modalities, strengthening complementary feature extraction. ExViT [46] innovatively combines a position-aware transformer with separable convolution to enhance multimodal data processing capability. HCT [47] constructs a collaborative framework of CNN and transformer to achieve multiscale feature learning from details to global. MACN [28] fuses self-attention and convolution operations to balance the global context and local detail capture. MSFMamba [52] explores the potential of SSMs in efficient sequence processing, providing a new technical paradigm for multimodal remote sensing data fusion.

Tables I–IV show the detailed classification performance of each method on four benchmark datasets. By analyzing the OA, AA, kappa, precision, recall, and $F1$ scores of different categories, it can be observed that MCAMamba leads existing methods in the vast majority of evaluation metrics, showing significant advantages especially in complex urban scene recognition and heterogeneous data fusion.

1) Results on the Houston2013 Dataset: Table I shows the classification performance of various methods on the Houston2013 dataset. MCAMamba achieved the highest OA of 94.75%, AA of 94.78%, kappa of 93.75%, precision of 95.49%, recall of 94.98%, and $F1$ score of 95.03%, demonstrating superior performance across all evaluation metrics. Its classification performance significantly outperforms other advanced methods in categories such as commercial (light cyan), road (green), highway (yellow), and parking lot2 (red). In Fig. 13, we fully filled the classification results of different methods and provided magnified views of key areas on the right side. It can be clearly observed that MCAMamba

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TABLE I
CLASS-WISE AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT FUSION METHODS ON THE HOUSTON 2013 DATASET

Class	FusAtNet	S^2ENet	ExViT	HCT	MACN	MSFMamba	MCAMamba
Health grass	81.38	81.48	79.33	80.63	81.39	**81.94**	81.38
Stressed grass	83.00	84.68	82.27	98.35	82.99	**99.82**	99.45
Synthetic grass	97.98	99.01	99.17	99.15	99.17	97.98	**99.19**
Trees	90.28	92.14	92.92	97.77	89.47	98.58	**99.19**
Soil	**100.00**	**100.00**	98.28	**100.00**	**100.00**	**100.00**	**100.00**
Water	98.68	**100.00**	88.16	97.37	97.37	98.68	98.68
Residential	76.24	**89.93**	86.73	87.95	88.91	82.57	84.36
Commercial	83.83	90.18	80.47	**96.45**	80.67	94.67	89.74
Road	87.13	92.63	81.72	87.50	**97.20**	94.03	96.08
Highway	61.49	64.86	73.87	54.81	59.92	69.35	**89.59**
Railway	**98.47**	97.60	90.87	96.12	96.50	92.67	98.45
Parking lot 1	97.19	88.09	97.38	**99.25**	98.13	96.44	98.88
Parking lot 2	93.06	91.93	88.89	90.28	87.50	**96.53**	88.19
Tennis court	99.06	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Running track	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA	87.57	88.35	87.75	90.74	88.90	92.19	**94.75**
AA	89.85	90.57	89.34	92.18	90.62	93.55	**94.78**
Kappa	86.50	87.37	86.69	89.95	87.94	91.50	**93.75**
Precision	92.60	90.02	91.12	92.61	92.04	92.97	**95.49**
Recall	89.85	90.57	89.34	92.18	90.62	93.55	**94.98**
F1	90.33	88.80	89.93	91.87	90.44	93.02	**95.03**

TABLE II
CLASS-WISE AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT FUSION METHODS ON THE HOUSTON 2018 DATASET

Class	FusAtNet	S^2ENet	ExViT	HCT	MACN	MSFMamba	MCAMamba
Health grass	86.95	**95.75**	94.54	94.87	80.68	87.96	94.80
Stressed grass	92.95	90.53	86.51	88.23	**94.94**	94.17	91.73
Artificial turf	99.88	**100.00**	**100.00**	**100.00**	98.93	**100.00**	**100.00**
Evergreen trees	99.15	99.42	98.16	99.02	98.57	99.34	**99.74**
Deciduous trees	98.12	99.21	98.48	98.87	99.20	98.86	**99.48**
Bare earth	98.11	99.99	99.94	99.99	96.76	99.94	**100.00**
Water	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Residential buildings	92.00	97.77	96.16	95.77	**98.27**	92.81	97.59
Non-residential buildings	93.38	93.63	92.44	94.45	93.39	95.13	**96.58**
Roads	64.55	75.25	72.69	69.30	74.67	78.55	**83.72**
Sidewalks	72.06	78.55	71.52	73.47	73.23	**81.07**	79.17
Crosswalks	93.46	96.90	87.47	95.64	94.22	92.98	**97.33**
Major thoroughfares	76.59	85.16	81.05	79.40	78.44	80.32	**86.41**
Highways	89.29	98.15	97.67	97.83	97.83	97.08	**98.74**
Railways	99.66	99.34	99.49	99.84	99.60	99.84	**99.91**
Paved parking lots	98.76	99.14	97.26	97.48	99.04	98.35	**98.84**
Unpaved parking lots	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
Cars	96.93	98.00	97.09	97.01	92.93	97.97	**98.62**
Trains	**100.00**	**100.00**	99.73	99.74	99.86	99.87	99.97
Stadium seats	99.95	**100.00**	99.90	99.98	99.81	99.91	**100.00**
OA	88.08	91.09	88.94	89.65	90.64	91.44	**93.35**
AA	92.59	95.34	93.51	94.04	94.82	94.71	**96.14**
Kappa	84.68	88.38	85.82	86.69	87.95	88.94	**91.48**
Precision	85.10	87.02	80.18	83.37	86.91	87.09	**87.13**
Recall	92.59	95.34	93.51	94.04	94.82	94.71	**96.13**
F1	87.04	89.96	84.88	86.98	89.36	90.07	**90.39**

makes precise delineation for commercial (light cyan), road (green), and healthy grass (dark blue) areas. Compared to other advanced models, MCAMamba has clearer category boundaries and correctly distinguishes the location information of roads and green spaces around commercial streets.

2) Results on the Houston2018 Dataset: Table II shows the classification performance of various methods on the Houston2018 dataset. MCAMamba achieved the highest OA of 93.35%, AA of 96.14%, kappa of 91.48%, precision of 87.13%, recall of 96.13%, and $F1$ score of 90.39%, demonstrating comprehensive performance superiority. Its classification performance significantly outperforms other advanced methods in categories such as residential buildings (gray), roads (red), railways (orange), and cars (pink). As shown in Fig. 14, the magnified area displays the ring road and central park area. MCAMamba can more accurately maintain the integrity and coherence of linear features such as roads (red), while demonstrating stronger discriminative ability for similar vegetation categories such as evergreen trees (green) and deciduous trees (dark green), effectively reducing category confusion phenomena.

3) Results on the Augsburg Dataset: Table III shows the classification performance of various methods on the Augsburg dataset. MCAMamba achieved the highest OA of 92.46%, AA

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TABLE III

CLASS-WISE AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT FUSION METHODS ON THE AUGSBURG DATASET

Class	FusAtNet	S^2ENet	ExViT	HCT	MACN	MSFMamba	MCAMamba
Forest	94.34	**98.92**	87.84	97.26	91.88	92.44	98.32
Residential area	92.56	**99.24**	95.81	98.63	96.36	98.85	97.71
Industrial area	47.70	11.19	31.93	36.61	51.13	27.98	**68.59**
Low plants	85.96	92.78	94.07	93.54	95.20	95.86	**96.64**
Allotment	49.33	51.44	41.02	51.43	27.65	**53.38**	50.74
Commercial area	10.17	0.98	**14.89**	7.97	8.41	4.01	1.74
Water	27.03	42.33	11.11	46.34	22.40	49.80	**57.22**
OA	87.52	89.82	87.04	88.47	89.56	91.33	**92.46**
AA	57.83	56.42	53.81	52.08	54.58	62.59	**67.34**
Kappa	81.85	84.84	81.10	83.05	83.36	86.63	**89.52**
Precision	60.53	75.48	57.07	61.77	59.38	68.70	**73.86**
Recall	57.83	56.42	53.81	52.08	54.58	61.19	**66.74**
F1	56.54	56.55	53.13	53.17	55.30	61.28	**68.69**

TABLE IV

CLASS-WISE AND OVERALL CLASSIFICATION ACCURACY (%) OF DIFFERENT FUSION METHODS ON THE BERLIN DATASET

Class	FusAtNet	S^2ENet	ExViT	HCT	MACN	MSFMamba	MCAMamba
Forest	76.17	83.59	80.81	77.81	81.71	83.58	**83.63**
Residential area	76.49	74.51	74.99	76.28	78.24	85.43	**90.22**
Industrial area	48.11	**66.35**	50.60	49.00	54.05	52.45	41.78
Low plants	71.25	63.27	74.35	**82.16**	80.36	67.77	71.25
Soil	74.96	91.67	87.86	87.34	**93.10**	76.81	78.28
Allotment	76.92	**80.18**	72.61	67.46	56.02	65.44	57.73
Commercial area	**28.86**	27.14	22.60	25.59	10.95	21.21	17.14
Water	46.45	35.64	45.12	42.47	40.35	53.34	**56.73**
OA	71.61	71.95	71.78	73.07	73.72	76.66	**79.18**
AA	62.40	**64.43**	63.62	63.51	61.85	63.75	62.07
Kappa	58.08	59.44	58.87	60.38	60.79	63.80	**66.73**
Precision	53.74	56.72	52.43	53.30	50.48	57.17	**60.39**
Recall	62.40	63.43	**63.62**	63.51	61.85	61.95	62.47
F1	55.56	58.38	55.49	56.62	54.19	58.67	**59.37**

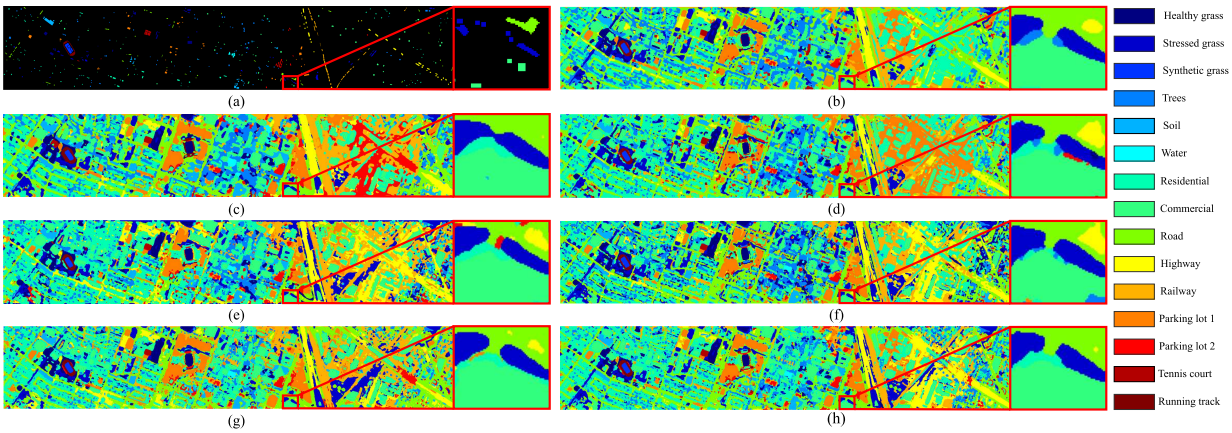


Fig. 13. Classification maps by different methods on the Houston2013 dataset. The right portion of each image shows a magnified view. (a) Ground truth. (b) FusAtNet (87.57%). (c) S^2ENet (88.35%). (d) ExViT (87.75%). (e) HCT (90.74%). (f) MACN (88.90%). (g) MSFMamba (92.19%). (h) MCAMamba (94.75%).

of 67.34%, kappa of 89.52%, precision of 73.86%, recall of 66.74%, and $F1$ score of 68.69%, showing consistent superiority across multiple evaluation metrics. It performs particularly well in categories such as industrial area (red) and water (purple). In Fig. 15, the interface between water and industrial areas is magnified. MCAMamba achieves more precise classification for the industrial area (red) and water (purple), two categories with insignificant differences in spectral features, with clear boundaries and high internal consistency, significantly outperforming other comparison methods.

4) Results on the Berlin Dataset: Table IV shows the classification performance of various methods on the Berlin dataset. MCAMamba achieved the highest OA of 79.18%, kappa of 66.73%, precision of 60.39%, and $F1$ score of 59.37%, demonstrating robust performance across key evaluation metrics. It performs well in categories such as residential

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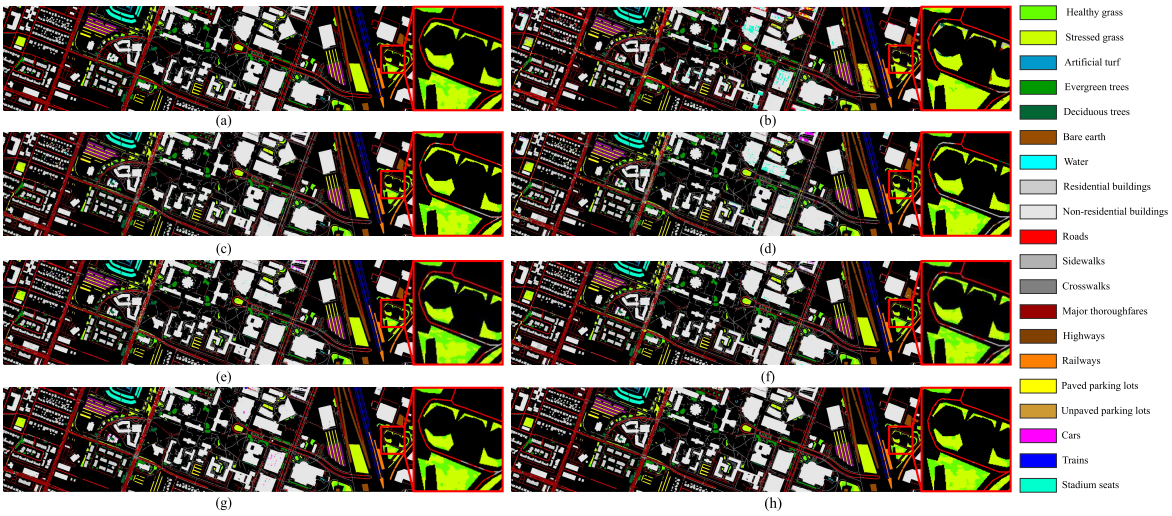


Fig. 14. Classification maps by different methods on the Houston2018 dataset. The right portion of each image shows a magnified view. (a) Ground truth. (b) FusAtNet (88.08%). (c) S^2ENet (91.09%). (d) ExViT (88.94%). (e) HCT (89.65%). (f) MACN (90.64%). (g) MSFMamba (91.44%). (h) MCAMamba (93.35%).

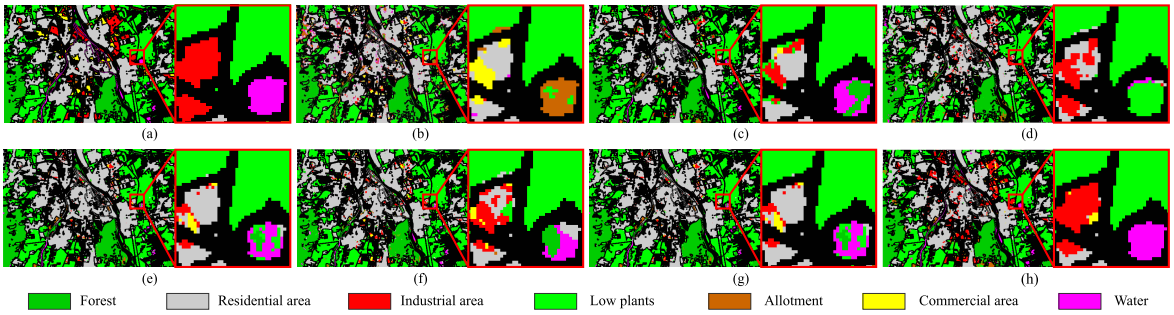


Fig. 15. Classification maps by different methods on the Augsburg dataset. The right portion of each image shows a magnified view. (a) Ground truth. (b) FusAtNet (87.52%). (c) S^2ENet (89.82%). (d) ExViT (87.04%). (e) HCT (88.47%). (f) MACN (89.56%). (g) MSFMamba (91.33%). (h) MCAMamba (92.46%).

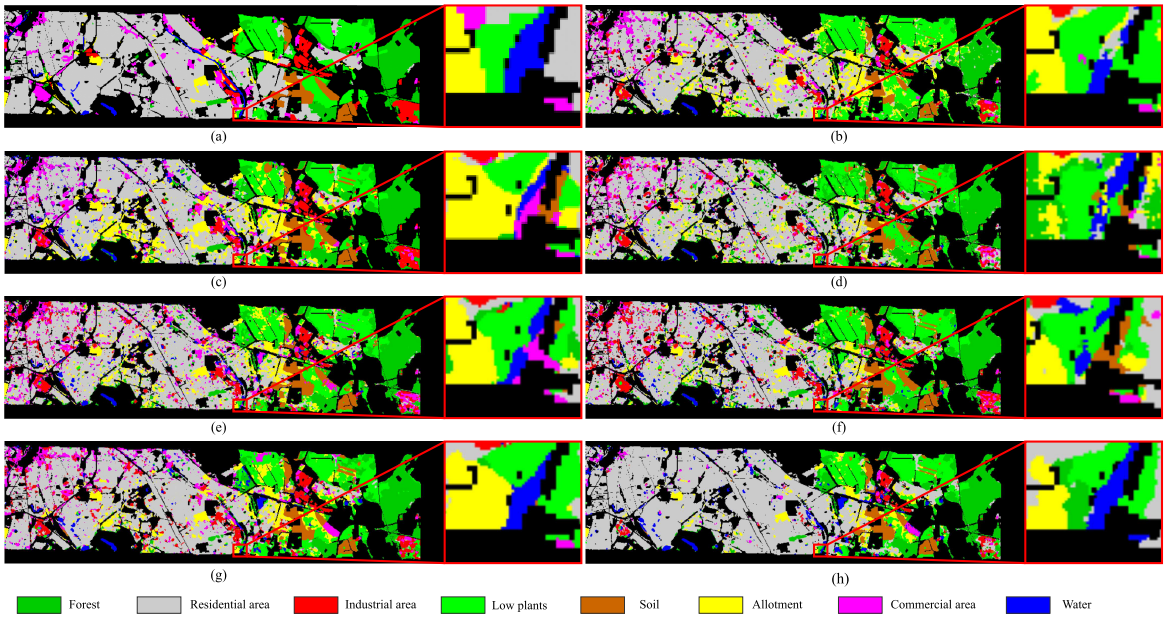


Fig. 16. Classification maps by different methods on the Berlin dataset. The right portion of each image shows a magnified view. (a) Ground truth. (b) FusAtNet (71.61%). (c) S^2ENet (71.95%). (d) ExViT (71.78%). (e) HCT (73.07%). (f) MACN (73.72%). (g) MSFMamba (76.66%). (h) MCAMamba (79.18%).

area (light gray) and water (blue). As shown in Fig. 16, the magnified area focuses on the river and its banks. MCAMamba provides more precise boundary delineation for water (blue), with the shoreline showing high coherence, while boundaries

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TABLE V
ABLATION STUDY ON DIFFERENT COMPONENTS OF THE MCAMAMBA MODEL

Components			Overall Accuracy (%)			
FExt	SSM	FFus	Houston2013	Houston2018	Augsburg	Berlin
✗	✗	✗	92.47	91.89	89.83	75.27
✓	✗	✗	93.42	92.06	89.99	75.32
✗	✓	✗	92.54	92.21	89.62	75.83
✗	✗	✓	93.42	92.03	90.57	75.41
✓	✓	✗	93.73	93.16	90.97	78.32
✓	✗	✓	93.93	92.64	91.25	77.36
✗	✓	✓	93.25	92.69	90.17	77.36
✓	✓	✓	**94.75**	**93.35**	**92.46**	**79.18**

between adjacent categories are clearer and confusion phenomena are significantly reduced.

5) Practical Implications: The experimental results from the four datasets fully validate MCAMamba's broad application potential. On the Houston2013 and Houston2018 datasets with HSI and LiDAR fusion, MCAMamba can precisely distinguish small-scale categories such as roads, railways, and cars, as well as precisely differentiate vegetation areas of different categories such as evergreen trees and deciduous trees. In the Augsburg and Berlin datasets with HSI and SAR fusion, it can better delineate boundaries for large-scale features such as industrial area, residential area, and water. MCAMamba demonstrates extensive practical application value, providing high-precision technical support for applications such as urban planning, transportation network analysis, and environmental monitoring.

D. Ablation Study

To systematically evaluate the effectiveness of MCAMamba's multilevel attention-guided SSM and its three-level feature fusion design, we conducted ablation experiments targeting each core module. We first removed all feature enhancement and fusion modules as the baseline model, then sequentially introduced FExt-Attention for single-modal feature enhancement, SSM-Attention for cross-modal interaction and sequence modeling, and FFus-Attention for multiscale feature fusion. Table V shows the classification accuracy of different module combinations on four benchmark datasets. The results show that the baseline model achieved OA of 92.47%, 91.99%, 89.83%, and 75.27% on the Houston2013, Houston2018, Augsburg, and Berlin datasets, respectively. Introducing FExt-Attention, SSM-Attention, or FFus-Attention individually all brought performance improvements, corresponding to enhancements in single-modal feature expression, heterogeneous modal interaction, and multiscale fusion capabilities. For example, FExt-Attention improved OA to 93.42%, 92.06%, 89.99%, and 75.32%. SSM-Attention showed particularly significant improvement in recognizing complex scenes in the Berlin dataset, with accuracy reaching 75.83%. FFus-Attention significantly enhanced the fusion effect on the Augsburg dataset, with accuracy reaching 90.57%.

Further introducing dual-module combinations validated the complementarity and synergistic effects between modules.

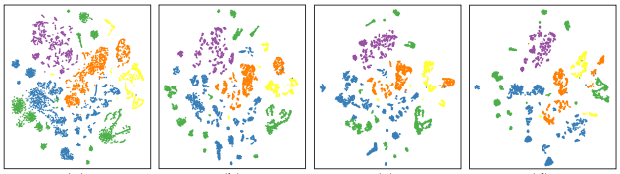


Fig. 17. t-SNE visualization of feature distributions with different components. (a) Baseline model. (b) Adding FExt-Attention. (c) Further adding SSM-Attention. (d) Complete MCAMamba model.

The combination of FExt-Attention and SSM-Attention achieved OA of 93.73%, 93.16%, 90.97%, and 78.32% on the four datasets. The combination of FExt-Attention and FFus-Attention reached 93.93% and 91.25% on the Houston2013 and Augsburg datasets, respectively. The combination of SSM-Attention and FFus-Attention improved OA to 92.69% on the Houston2018 dataset.

When all three core modules worked collaboratively, MCAMamba achieved the best performance on all datasets, with OA of 94.75%, 93.35%, 92.46%, and 79.18%. These results fully validate the effectiveness of deep fusion between multilevel attention and SSMs, significantly enhancing the heterogeneity modeling and information synergy capabilities of multisource remote sensing data.

We progressively introduced modules to MCAMamba on the Houston2013 dataset and conducted t-SNE visualization analysis of its feature representations. t-SNE effectively reveals the inter-class separation and intra-class aggregation of feature distributions by mapping high-dimensional features to a 2-D space [71]. As shown in Fig. 17, the baseline model (a) exhibits obvious category overlap in feature distribution with low discriminability. After introducing the FExt-Attention module for single-modal feature enhancement (b), preliminary feature clustering structures form with improved category separation. Further adding SSM-Attention to achieve cross-modal interaction and sequence modeling (c) results in clearer inter-class boundaries and significantly enhanced clustering density. Finally, after adding FFus-Attention for multiscale feature fusion to form the complete MCAMamba model (d), the feature distribution of each category exhibits highly compact intra-class structure and clear inter-class boundaries. These visualization results intuitively validate the progressive enhancement effect of MCAMamba's multilevel attention, further demonstrating the effectiveness of each module in improving the feature discriminability and clustering of multisource remote sensing data.

E. Category-Wise Module Contribution Analysis

To deeply analyze the relationship between MCAMamba's superior performance on different classes and its module design, we conducted detailed category-wise ablation studies on the Houston2018 dataset with 20 categories. Table VI presents the OA of MCAMamba and when key modules are removed for each land cover class. We found differences in module effectiveness across different categories. For vegetation classes (health grass, stressed grass, evergreen trees, and deciduous trees), FExt-Attention shows the most significant contribution, with improvements of 1.63%, 0.46%, 0.24%, and

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TABLE VI
CATEGORY-WISE ABLATION STUDY: MODULE CONTRIBUTION ANALYSIS ON HOUSTON2018 DATASET (%)

Class	Individual Module Performance			Full Model	Module Contribution		
	w/o FExt	w/o SSM	w/o FFus		ΔFExt	ΔSSM	ΔFFus
Health grass	93.17	94.45	94.35	94.80	**1.63**	0.35	0.45
Stressed grass	91.27	91.72	91.06	91.73	0.46	0.01	**0.67**
Artificial turf	100.00	100.00	100.00	100.00	0.00	0.00	0.00
Evergreen trees	99.50	99.25	99.63	99.74	0.24	**0.49**	0.11
Deciduous trees	99.42	99.20	99.48	99.48	0.06	**0.28**	0.00
Bare earth	99.88	99.96	99.98	100.00	**0.12**	0.04	0.02
Water	100.00	99.64	100.00	100.00	0.00	**0.36**	0.00
Residential buildings	97.24	97.48	97.55	97.59	**0.35**	0.11	0.04
Non-residential buildings	96.34	95.51	96.20	96.58	0.24	**1.07**	0.38
Roads	81.35	83.54	82.38	83.72	**2.37**	0.18	1.34
Sidewalks	76.15	76.00	79.21	79.17	**3.02**	3.17	-0.04
Crosswalks	97.17	95.09	96.18	97.33	0.16	**2.24**	1.15
Major thoroughfares	83.77	86.06	84.91	86.41	**2.64**	0.35	1.50
Highways	98.71	97.86	97.42	98.74	0.03	0.88	**1.32**
Railways	99.86	99.87	99.71	99.91	0.05	0.04	**0.20**
Paved parking lots	98.80	98.44	98.29	98.84	0.04	0.40	**0.55**
Unpaved parking lots	100.00	100.00	100.00	100.00	0.00	0.00	0.00
Cars	97.95	98.42	98.58	98.62	**0.67**	0.20	0.04
Trains	99.79	99.24	99.82	99.97	0.18	**0.73**	0.15
Stadium seats	100.00	100.00	100.00	100.00	0.00	0.00	0.00
Overall Accuracy	**92.63**	**92.62**	**93.00**	**93.39**	**0.76**	**0.77**	**0.39**

TABLE VII
COMPARISON OF DIFFERENT MODEL ARCHITECTURES ACROSS DATASETS

Model Architecture	Houston2013	Houston2018	Augsburg	Berlin
OnlyMamba	93.17	92.58	90.64	73.80
FullAttn	93.48	93.12	91.15	74.09
MambaAttn	93.92	92.88	91.54	78.06
MCAMamba	**94.93**	**93.34**	**92.43**	**79.11**

0.06%, respectively. This indicates that parallel spatial-Cha-Attn is particularly effective for enhancing spectral feature discrimination in vegetation recognition. For infrastructure classes with complex spatial structures (roads, sidewalks, and major thoroughfares), FExt-Attention again provides the largest gains (2.37%, 3.02%, and 2.64%), demonstrating its capability in capturing spatial morphological features. In contrast, for highly distinctive linear features (crosswalks and railways), SSM-Attention shows superior contribution (2.24% and 0.73%), indicating that cross-modal sequence modeling excels at recognizing structured patterns. Interestingly, for transportation-related classes (highways and paved parking lots), FFus-Attention contributes most significantly (1.32% and 0.55%), suggesting that multiscale feature fusion is crucial for distinguishing similar urban infrastructure. These category-specific insights validate that MCAMamba's three-level design addresses different classification challenges through specialized module contributions.

F. Model Architecture Analysis

Based on the same baseline framework, this study designed four architectural variants for comparative analysis, as shown in Table VII. OnlyMamba adopts a pure SSM structure, extracting initial features through 3-D/2-D convolutions followed by Mamba block processing and simple concatenation, achieving OA of 93.17%, 92.58%, 90.64%, and 73.80%, computationally efficient but lacking dedicated spatial feature extraction and modal interaction mechanisms. FullAttn is built on multilevel attention, employing parallel spatial-Cha-Attn in the feature extraction stage, multihead self-attention in the modal interaction stage, and combining cascaded attention with global context in the feature fusion stage, achieving OA of 93.48%, 93.12%, 91.15%, and 74.09%, with improved spatial features and modal fusion but higher computational complexity. MambaAttn selectively integrates attention with Mamba, using attention only in the feature extraction and simplified fusion stages, achieving OA of 93.92%, 92.88%, 91.54%, and 78.06%, showing outstanding performance in balancing computational efficiency with feature extraction capability. MCAMamba, as a complete hybrid architecture, systematically integrates attention and SSMs across all three stages, improving OA to 94.75%, 93.35%, 92.46%, and 79.18%, achieving full-process collaborative processing of single-modal feature enhancement, heterogeneous data fusion, and feature optimization.

Compared to OnlyMamba, FullAttn, and MambaAttn, MCAMamba's average OA improved by 2.39%, 1.98%, and 0.84% points, respectively. The experimental results indicate that as the depth of integration between attention and SSMs increases, classification performance progressively improves, validating the complementarity of these two technologies in processing multisource remote sensing data. Through differentiated design across three key stages, MCAMamba effectively addresses the heterogeneity and high-dimensional characteristics of multisource remote sensing data, providing an architectural solution that balances both performance and efficiency for remote sensing image classification.

G. Analysis of Multiscale Spatial-Cha-Attn Effects

The multiscale spatial-Cha-Attn (MSC-Attn-P) module is a key component in MCAMamba for enhancing single-modal feature expression capabilities. This module processes HSI

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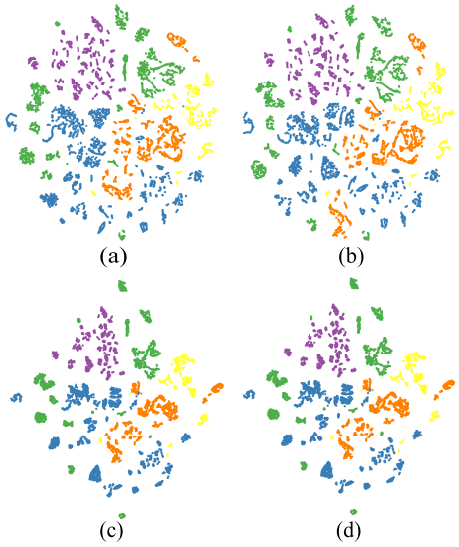


Fig. 18. t-SNE visualization of HSI and LiDAR/SAR features on the Houston2013 dataset before and after multiscale spatial-Cha-Attn. (a) HSI features without MSC-Attn-P. (b) SAR features without MSC-Attn-P. (c) HSI features with MSC-Attn-P. (d) SAR features with MSC-Attn-P.

and LiDAR/SAR data independently, fully exploiting the characteristics of each modality: HSI focuses on strengthening spectral discriminative information, LiDAR highlights elevation structure, and SAR enhances scattering and spatial texture. By analyzing the role of MSC-Attn-P in the FExt-Attention module, we can intuitively evaluate the contribution of single-modal feature enhancement to the fusion effect. Fig. 18 shows the t-SNE visualization results on the Houston2013 dataset, with (a) and (b) showing the HSI and LiDAR/SAR feature distributions without applying MSC-Attn-P, and (c) and (d) showing the distributions after enhancement.

The visualization results show that the original convolution features exhibit category overlapping, with particularly low inter-class separation in LiDAR/SAR data. After introducing MSC-Attn-P, the discriminability and clustering of HSI features significantly improve, and the structural and texture information of LiDAR/SAR is also effectively enhanced. Through modality-adaptive design, MSC-Attn-P emphasizes Cha-Attn for HSI and Spa-Attn for LiDAR/SAR, with both working in parallel coordination, significantly improving the foundation for fusion. This validates the effectiveness of MCAMamba's staged feature optimization strategy.

H. Computational Efficiency Analysis of Mamba Versus Transformer for Long Sequence Processing

To validate the advantages of Mamba over Transformer in long-sequence modeling and computational efficiency, we constructed a comparative experiment. Specifically, we replaced the core Mamba architecture in MCAMamba with an equivalent transformer architecture while keeping other modules unchanged, constructing the MCATransformer model. Under the same hardware environment, we adjusted the Winsize to control the input sequence length and tested the computational performance of both architectures under different sequence length conditions.

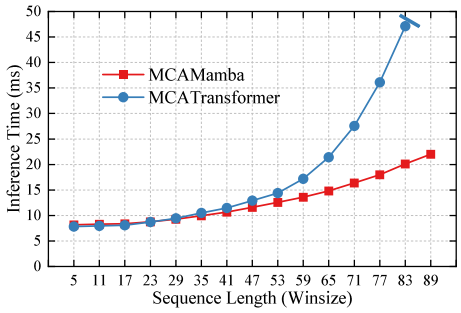


Fig. 19. Comparison of inference time between MCAMamba and MCATransformer under different window sizes. As sequence length increases, MCATransformer exhibits quadratic complexity growth while MCAMamba maintains linear complexity growth.

As shown in Fig. 19, with the incremental increase of window size, MCAMamba and MCATransformer exhibit significant performance differences in inference time. Within the shorter sequence length range (Winsize 5–23), both architectures demonstrate comparable inference times. However, as sequence length continues to grow, the transformer architecture exhibits typical quadratic complexity growth characteristics in inference time, while the Mamba architecture maintains approximately linear growth. The experimental results are highly consistent with theoretical analysis. When Winsize reaches 83, MCATransformer's inference time increases to 47.8 ms, while MCAMamba requires only 20.5 ms. More importantly, when Winsize exceeds 83, MCATransformer cannot continue running due to memory limitations, while MCAMamba can still stably process longer sequences.

This experimental result powerfully confirms the computational advantages of Mamba architecture in long-sequence data processing. For remote sensing image classification tasks, Mamba's linear complexity characteristics enable it to handle larger image sizes and higher resolution data, significantly reducing computational overhead and providing a more feasible technical pathway for large-scale remote sensing data processing in practical applications.

V. Conclusion and Future Work

This article introduced MCAMamba, a novel architecture designed for multisource remote sensing joint classification. It innovatively adopts an attention-guided Mamba systematic integration model, with an innovative design of a three-level feature fusion architecture: the FExt-Attention module collaboratively enhances the rich spectral information of HSI and the structural features of LiDAR/SAR through parallel spatial-Cha-Attn. The SSM-Attention module fuses Cross-Attn with SSMs to achieve deep interaction of heterogeneous data. The FFus-Attention module integrates global context with cascaded spatial-Cha-Attn to achieve adaptive fusion of multiscale features. Experiments on four public datasets, Houston2013, Houston2018, Augsburg, and Berlin, demonstrate that MCAMamba achieves overall classification accuracies of 94.75%, 93.35%, 92.46%, and 79.18%, reaching a state-of-the-art level.

Despite MCAMamba's excellent performance, there remains room for optimization in computational efficiency

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and classification of certain complex categories. Future research will focus on theoretical foundations and application extensions: 1) exploring lightweight attention and SSM collaborative mechanisms to improve computational efficiency; 2) researching adaptive parameter optimization strategies to enhance adaptability to different scenarios; 3) expanding support for the fusion of more types of remote sensing data sources; 4) combining explainable AI technologies to improve decision transparency. These improvements will further leverage MCAMamba's application potential in the field of Earth observation, providing reliable support for intelligent interpretation of multisource remote sensing data.

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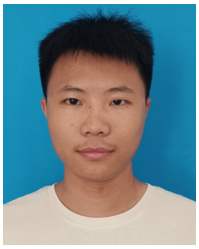
Mingyu Dou received the B.S. degree from Xi'an University of Technology, Xi'an, China, in 2024. He is currently pursuing the M.S. degree with the University of Chinese Academy of Sciences (UCAS), Beijing, China.

His research interests include multisource data fusion, image semantic segmentation, open-vocabulary change detection, water-related optics, and biomedical signal processing.



Shi Qiu received the M.S. degree from Xidian University, Xi'an, China, in 2012, and the Ph.D. degree from the University of Chinese Academy of Sciences, Xi'an, in 2016.

He is Associate Research Fellow with Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences. He is a New Star of science and technology at Shaanxi and a Young Scholar at Western China. He has authored or co-authored more than 50 SCI papers. His research interests include image restoration, sparse representation, and object detection.



Ming Hu received the B.S. degree from Xi'an University of Technology, Xi'an, China, in 2024. He is currently pursuing the M.S. degree with the University of Chinese Academy of Sciences, Beijing, China.

His research interests include data-centric approaches for intelligent analysis, covering semantic segmentation, semi-supervised learning, biomedical signal processing, and zero-shot anomaly detection.

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Xiaozhen Qiao received the bachelor's degree from China University of Petroleum (East China), Beijing, China, in 2024. He is currently pursuing the Ph.D. degree with the School of Information Science and Technology, University of Science and Technology of China (USTC), Hefei, China, under the supervision of Prof. Xuelong Li.

His research interests primarily focus on vision–language models and 3-D reconstruction.



Huping Ye received the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2017.

He is an Associate Professor at the Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences. His research interests include uncrewed aerial vehicles (UAVs) remote sensing data process, UAV air route, water color remote sensing and water optics, mainly including remote sensing inversion of water quality parameters, and inherent optical properties measurement and inversion.



Xiaohan Liao (Member, IEEE) received the Ph.D. degree in geography from Oxford University, Oxford, England, in 1992.

From 1988 to 1997, he researched on climate change simulation and atmospheric remote sensing applications in the Goddard institute of space studies, NASA, Washington, DC, USA. At present, he is engaged in the research about remote sensing applications for light and small uncrewed aerial vehicles (UAVs), UAV remote sensing mega data carrier, and networking observation technology, cloud-end control, and low-altitude air route construction for UAVs with Chinese Academy of Sciences, Beijing, China.

Dr. Liao serves as a Supervisory of China Association for Geographic Information Society and the Chair of the Committee for UAV Applications and Regulation. He is the Chair of the IEEE SA 1939.1 Working Group–Standard for a Framework for Structuring Low Altitude Airspace for UAV Operations.



Zhe Sun received the Ph.D. degree from Beijing University of Technology, Beijing, China, in 2014.

He has been an Associate Professor at Northwestern Polytechnical University, Shenzhen, China, since 2022. Before that, he was a Post-Doctoral at Friedrich Schiller University, Jena, Germany, from 2020 to 2022, and Helmholtz Institute, Jena, GSI Helmholtzzentrum fuer Schwerionenforschung GmbH, Darmstadt, Germany, from 2018 to 2020. Previously, he contributed as a Research Assistant at Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, from 2014 to 2018. His research interests include water-related optics, computational imaging, and laser imaging.

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